

or sometimes even to the code’s programmer in the future. Almost every programmer has asked themselves, “Who wrote this unreadable mess?” only to find the answer is, “I did.”

Docstrings are a Python-specific form of documentation for your functions, methods, and modules. When you specify comments in the docstring format, automated tools, such as documentation generators or Python’s built-in `help()` module, make it easy for developers to find information about your code.

Type hints are directives you can add to your Python source code to specify the data types of variables, parameters, and return values. This allows static code analysis tools to verify that your code won’t generate any exceptions due to incorrectly typed values. Type hints first appeared in Python 3.5, but because they’re based on comments, you can use them in any Python version.

This chapter focuses on the aforementioned three techniques for embedding documentation inside your code to make it more readable. External documentation, such as user manuals, online tutorials, and reference materials, are important but not covered in this book. If you want to learn more about external documentation, check out the Sphinx documentation generator at <https://www.sphinx-doc.org/>.

Comments

Like most programming languages, Python supports single-line comments and multiline comments. Any text that appears between the number sign `#` and the end of the line is a single-line comment. Although Python doesn’t have a dedicated syntax for multiline comments, a triple-quotes multiline string can serve as one. After all, a string value by itself doesn’t cause the Python interpreter to do anything. Look at this example:

```
# This is a single-line comment.

"""This is a
multiline string that
also works as a multiline comment. """
```
